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Advancing Geothermal Well Engineering Through Cutting-Edge Drilling and Workover Designs

J. J. R. Cayao, Welltec A/S, Denmark; R. E. S. Divinagracia, Energy Development Corporation, Philippines

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Abstract

The distinct characteristics of geothermal wells necessitate a unique and specialized approach to well design. This is critical for prolonging well lifespan and tackling the unique operational challenges present in mature geothermal fields, such as the case on a field in the Philippines. To enhance steam production, geothermal wells often intersect faults and complex geological formations, which result to partial to total fluid losses. These losses introduce significant uncertainty in cementing and compromise its quality.

Such issues can lead to channeling, the formation of micro-annuli, and cement degradation — all of which undermine well integrity and reduce production efficiency. Furthermore, these cementing challenges, especially in the case of geothermal wells, may result to having trapped fluid in casing-to-casing annulus that can potentially compromise well integrity due to huge temperature swings that the wells encounter over their operating lives. Addressing these integrity concerns in a cost-effective manner is critical to maintaining safe, efficient, and economically viable well operations over time.

To overcome these challenges, Welltec partnered with a geothermal operator in the Philippines to rehabilitate eight geothermal production wells affected by casing collapse and parted casing. A traditional workover involving bridge plugs was deemed impractical, as drilling out the plug risked further damage to liners exposed to corrosive, low-pH fluids. There was also a concern that trapped fluids could trigger additional casing failures. Therefore, a more controlled and precise cementing strategy was required to restore casing integrity.

This paper introduces a tailored solution that integrates high-temperature Metal Expandable Packer technology and stage cementing collars. Case studies from a geothermal field in the Philippines illustrate how this approach restores well integrity without the need for a dedicated drill-out run.

Building on experience from prior deployments, the improved MEP enables long-string tieback and casing relining to depths of approximately 1,300 meters — previously considered unattainable due to casing collapse, which had limited operations to around 700 meters.

New well designs using CRA are also being optimized to benefit from such solution and withstand highly acidic environments. These wells require immediate relining, applying insights from earlier CRA wells in the same field. The MEP system eliminates the need to drill out plugs, thereby preventing damage to CRA casing.

Integrating MEP technology into both drilling and workover operations offers a robust strategy to resolve integrity issues that typically lead to early well failure. This geothermal-specific technology enhances repeatability, safety, and operational control, ultimately enabling more reliable and extended utilization of geothermal resources.

Introduction

The engineering and operational complexities associated with the design, construction, and maintenance of geothermal wells are profound, particularly in high-acidity geothermal reservoirs where environmental and operational conditions impose severe constraints on well integrity (Ng'ang'a and Maleche, 2017). Such environments are characterized by extreme pH levels, often descending to approximately three, which catalyze aggressive corrosion processes, thereby jeopardizing the structural integrity of well casings, liners, and associated well components throughout the lifecycle of the well. These challenges are further exacerbated by the demanding geological formations encountered in geothermal fields, which necessitate sophisticated drilling and workover strategies and robust tools capable of enduring thermal stress, corrosive fluids, and high-pressure conditions.

Recent studies, including the one of Jolie et al. (2021), emphasize that optimizing geothermal well performance particularly the maximization of sustainable steam recovery necessitates the deployment of well designs engineered explicitly for integrity preservation. Failures in well integrity not only compromise operational safety but also introduce substantial environmental risks, such as surface contamination and fluid leakage, which in turn elevate operational expenditures and diminish overall energy recovery efficiency. In geothermal systems where pH levels are critically low, traditional well completion and intervention technologies often fall short, resulting in expedited casing degradation, frequent workover interventions, and costly repairs.

In the context of the Philippines, a region with abundant geothermal resources, the operational landscape is further complicated by highly acidic geothermal fluids, which impose significant corrosion burdens on well infrastructure. In one specific geothermal field, wells are subjected to stresses from the need of multiple well stimulations to induce well discharge and high temperature deltas between producing and quenched states. The typical operational challenges include rapid deterioration of casing materials, formation of corrosion-induced or annular pressure buildup-induced wellbore irregularities, and recurrent failures that necessitate costly workovers (Jimenez, 2018). Meanwhile, there exists a conspicuous absence of durable, high-temperature resistant well completion technologies capable of mitigating these pervasive integrity issues. This scenario underscores the urgent need for innovative solutions that can withstand aggressive environmental conditions while maintaining wellbore stability and operational efficiency.

This study explored the deployment of advanced Metal Expandable Packers (MEPs), engineered specifically for high-temperature, corrosive environments, as a potential remedial and preventative measure for well integrity enhancement. The investigation details the application of a novel MEP design integrated with a cement port collar in conjunction with a specialized reline or tieback casing string, aimed at facilitating long-term well integrity and reducing operational costs associated with maintenance and repair. The proposed configuration demonstrated an effective technique for conducting long well relining and tieback operations, thus reducing the risk of further damage to existing liners.

According to New Zealand's Code of Practice for Deep Geothermal Wells (NZS 2403:2015 (2015), critical to this approach is the enhancement of cementing quality through improved seal integrity and zonal isolation, which serve to restore or reinforce compromised casing intervals, specifically targeting the 13-3/8 in. and 9-5/8 in. production casing sections. The combination of these technological innovations provides a robust framework for mitigating corrosion-induced degradation and milled casing intervals from annular pressure buildup-induced wellbore irregularities, extending well lifespan, and facilitating more efficient, less intrusive workover operations.

Methodology and Solution Design

Relining with cementing is usually undertaken to address casing problems in the 9-5/8 in. production casing in instances of regular-sized holes and 13-3/8 in. casing in instances of big-hole casing configurations. The conventional methodology involves deploying a drillable bridge plug, which is positioned around the top or near the existing liner to serve as a temporary isolation barrier during the subsequent relining or tieback casing and cementing operation. This bridge plug, being non-retrievable after setting, necessitates subsequent drilling-out to remove it from the wellbore, alongside the challenge of debris removal down to the perforated liner or production zone. Such procedures inherently pose significant risks to existing liners, particularly given their compromised condition and the challenging environment, which includes exposure to corrosive fluids, thermal fatigue, and mechanical stresses. The potential for debris entrapment, incomplete removal, or damage to the liner during drill-out presents operational hazards such as compromised zonal isolation, increased non-productive time, and jeopardized well integrity.

The beginning of the long relining concept, especially through the utilization of Metal Expandable Packers (MEPs) combined with port collars, signifies a transformative approach to long-relining and deep production casing repairs. This technique was developed to address the technical and economic constraints associated with traditional relining methods in wells with deep production sections. By integrating MEPs and port collars, this approach eliminates the need to have long liner laps to be able to reline shallower at around 700m to 800m, which affect not only the cost, but also the overall performance and life of the well.

However, as this solution aimed to address the need for a long relining solution without a dedicated drill-out run, this solution also brought forth challenges, mainly pertaining to having a good reline cementing job. The high delta temperature from the MEP setting depth, the high hydrostatic pressure of cement slurry that poses risks to the casing integrity in terms of the burst and collapse rating, as well as the capacity of the MEP needs to be addressed to make this concept viable for operation.

These challenges were all addressed by the bespoke MEP design developed for geothermal application. Only through addressing these multifaceted technical challenges can the long relining strategy be made operationally viable, economically advantageous, and capable of delivering confidence in well integrity restoration for deep, high-risk geothermal well applications.

Solution Design for Low pH, High Temperature Geothermal Wells

The minimum reline setting depth or "critical depth" is determined by the expected reservoir pressure. However, the current workover design on low pH wells is venturing on long relines to ensure that the whole production casing will be repaired instead of just being slightly below the minimum reline setting/critical depth. In geothermal operations, the cost and complexity of drilling new wells or executing major workovers are significant, particularly when well integrity is at risk in these ultra-high temperature wells. Furthermore, with high thermal cycling, where wells temperature can fluctuate from ~40 °C to over 320 °C, imposes extreme stress on tubulars and cement systems. In such an environment, conventional cementing alone often falls short in delivering long-term zonal isolation. Trapped annular fluids can also lead to sustained casing pressure, casing deformation, or catastrophic collapse that would significantly shorten a lifespan of a well that is producing a renewable energy resourced.

To address the challenges posed by extreme conditions in geothermal environment, an advanced Metal Expandable Packer (MEP) specifically engineered for geothermal well construction and intervention was developed as shown in [Figure 1](#). Unlike traditional barriers, the MEP employs a hydraulically actuated metal sleeve integrated with the liner or casing, forming a rapid and permanent annular seal within minutes of activation. The system's design allows full rotation and circulation during deployment, ensuring maximum reach and placement accuracy without risk of premature activation which is critical for challenging well paths and re-entry scenarios.



Figure 1—Specifically Engineered Metal Expandable Packer (MEP): (a) Ultra High Temperature MEP; (b) High Temperature MEP. (Cayao, 2025)

While this MEP technology has an extensive track record in oil and gas, its adaptation for geothermal pushes the technology into harsher operational domains specifically in terms of thermal stress. In prior geothermal applications, the MEP has been proven as both a primary mechanical barrier and a secondary sealing element to complement cement, delivering redundancy and integrity in high-temperature, highly corrosive environments. Earlier used cases were limited to shallow relines (<780 mMD), but recent enhancements in metallurgical resilience and thermal sealing capability now enable deployment at significantly greater depths.

As illustrated in [Figure 2](#), representative well profiles employing MEP technology demonstrate advanced capabilities in casing relining and tieback operations within geothermal wells. In [Figure 2a](#), the MEP is strategically positioned on the tail pipe, with a cement port collar situated above it. This configuration facilitates the achieving of comprehensive casing cementing, enabling the effective reline of extended casings which significantly exceeding the limitations of earlier shallow relines. This solution design enhances the integrity and durability of the reline by ensuring a uniform, fully cemented casing annulus over a length that was previously challenging to attain while maintaining well integrity through elimination of sustained casing pressure (SCP) that is highly detrimental for geothermal wells.

Conversely, [Figure 2b](#) highlights the versatility of the same MEP design, emphasizing its robustness and adaptability for multiple functions, including casing tieback and zonal isolation in open-hole sections for well construction. The inherent flexibility of the design allows it to be deployed in both cased and open hole environments without the need for significant modification. This multifunctionality is achieved through the ability of the MEP to conform to varying borehole geometries, accommodating irregularities such as casing ovality and borehole washouts. The expandable features of the MEP, capable of conforming to borehole diameters up to 14 in. for the 1214MEP and 10 in. for the 812MEP, further optimize their functionality in challenging geological conditions. This bespoke configuration thereby integrates a high degree of adaptability, enabling the same MEP system to efficiently perform in diverse operational scenarios ranging from casing installation and reinforcement within cased holes to zonal isolation in open hole sections. This reduced the need for multiple specialized tools and streamlined deployment procedures.

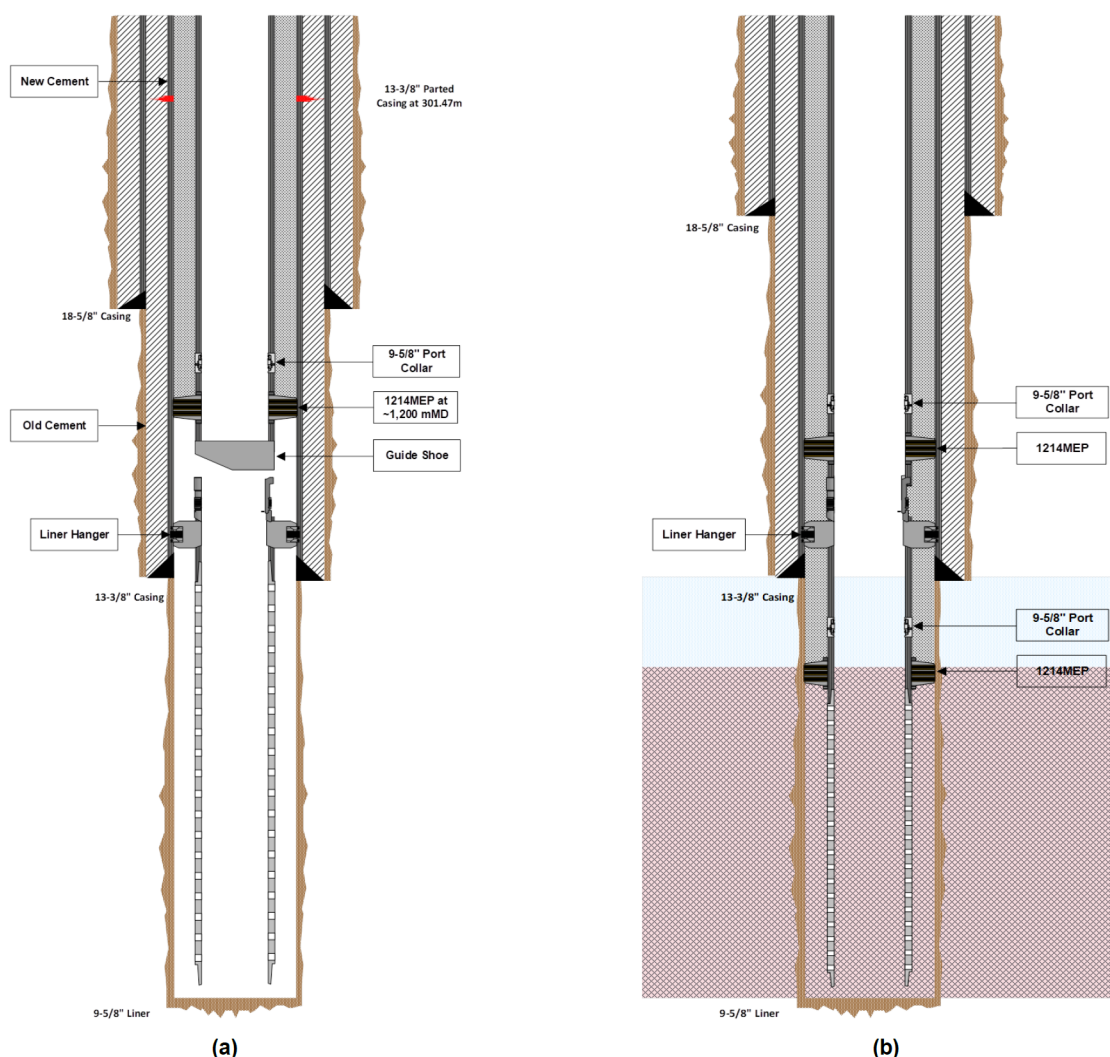


Figure 2—Geothermal Well Profile Utilizing MEP Technology: (a) for Workover Operations; (b) for Drilling Operations (Cayao, 2025)

Technical Benefits and Engineering Controls for Operational Risks

The incorporation of Metal Expandable Packer (MEP) technology into geothermal well remediation strategies offer significant advancements in maintaining and restoring well integrity over the life of the well, particularly in ultra-high temperature and corrosive downhole environments. The MEP functions as a permanent, high-integrity mechanical barrier capable of withstanding temperatures exceeding 300 °C, making it ideally suited for applications in high-enthalpy geothermal reservoirs. By eliminating the need for a dedicated drill-out run during well relining, the deployment of thermally robust MEPs contributes to reduced rig time and lower overall intervention costs.

To further support cement placement and achieve full zonal coverage, a two-stage cement port collar is utilized and strategically positioned above the MEP. While this configuration enhances cementing efficiency, simulations conducted during the design phase revealed a critical risk of fluid entrapment between the cement port collar and the MEP. This trapped fluid, if thermally expanded, could exert considerable stress on the relined casing even though the MEP is rated to withstand differential pressures up to 6,000 psi. To mitigate this risk, rupture disks are incorporated between the MEP and the cement port collar, functioning as controlled pressure-relief devices. This design enhancement provides a margin of safety and greater control over pressure behavior within the annular space during thermal cycling.

Additionally, a valve block system detailed in Figure 3 is integrated into the MEP assembly to accommodate elevated differential pressures and to serve as a secondary pressure mitigation mechanism. In conjunction with the two rupture disks placed on the base pipe above the MEP expansion sleeve, an additional rupture disk is embedded within the valve block itself. This design allows any trapped fluid between the base pipe and the MEP's expansion sleeve to be safely vented, thereby protecting both the tool and the surrounding casing. With these integrated pressure management features and the MEP's mechanical capacity to hold high differential loads, long-section relining operations, ranging from approximately 1,000 m TVD to 2,200 m TVD, are considered technically feasible and operationally robust.

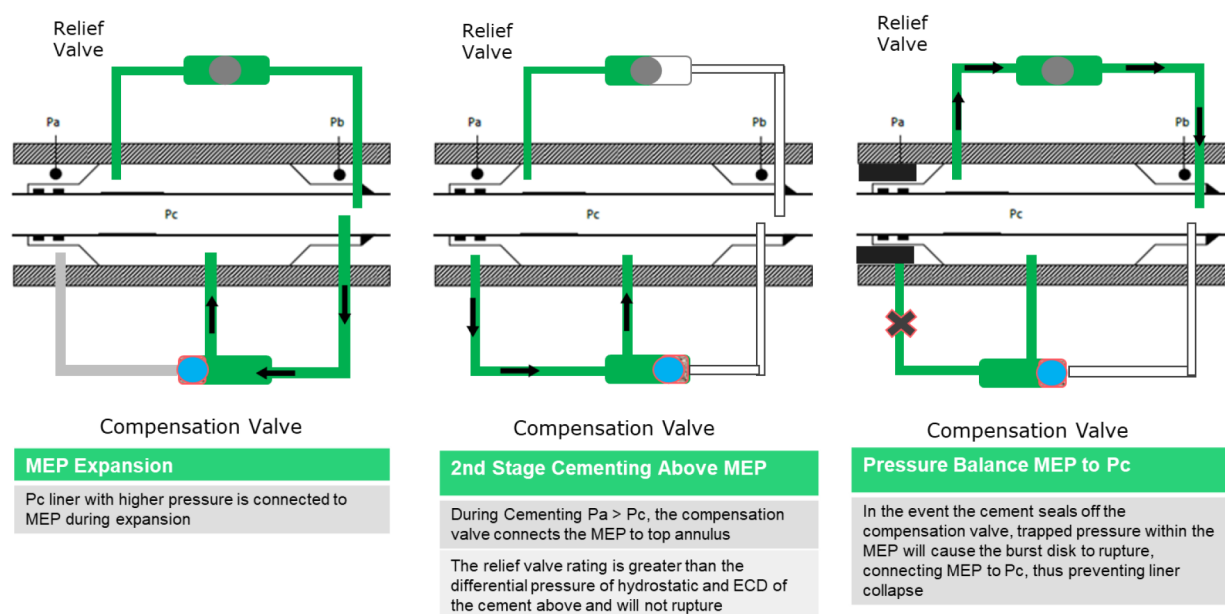


Figure 3—Valve block configuration for geothermal applications to address long relining and risk of trapped fluid (Cayao, 2025)

While these engineering advancements offer substantial performance improvements in geothermal well remediation and construction, the deployment of MEP technology is not without operational risk, particularly in wells characterized by severe casing degradation. Casing conditions play a pivotal role in the success or failure of MEP deployment and remain one of the primary concerns when planning and executing relining operations.

In many geothermal wells, particularly those subjected to extended production cycles and exposure to chemically aggressive fluids, casing integrity can be significantly compromised. Localized corrosion, extensive wall thinning, mechanical deformation, collapse, and parting of casing are frequently observed in both production and injection wells, especially in the presence of low pH fluids, high-chloride brines, and abrasive silica-laden formations.

From the standpoint of deployment planning, the operational flexibility of MEPs facilitates their applications across a wide range of wellbore conditions, including both newly drilled geothermal wells and legacy wells undergoing workover. In recently completed wells where well integrity of the casing string has been verified, deployment can proceed with minimal preparatory casing diagnostics. For these scenarios, tieback liners or relining operations incorporating MEPs can be deployed efficiently, minimizing rig time and intervention costs.

Conversely, for geothermal wells which were long exposed to aggressive downhole chemistries such as high chloride concentrations, elevated temperatures, abrasive fluids, or low pH fluids, it becomes critical to perform comprehensive well integrity assessments. Caliper logging is especially recommended in these corrosive or erosive settings to detect casing ovality, internal diameter reductions, or scale deposition, all

of which could compromise the solution offered by a bespoke MEP technology due to potential channeling of fluid behind the aforementioned packer. Furthermore, the utilization of multi-arm caliper tools, in conjunction with pressure-temperature-spinner (PTS) surveys, enables high-resolution mapping of casing wear, eccentricity, and possible leak paths due to casing thinning or microannular flow which are very prevalent in these highly erosive and corrosive wells.

Nonetheless, a notable advantage of MEPs in geothermal workover operations is their radial expansion capability, which allows them to conform to variable borehole geometries. This attribute is particularly valuable in damaged casings with non-uniform ID and geometry, as well as open hole sections where prior casing and cementing programs have failed to provide effective zonal isolation often due to poor cement placement, thermal cycling-induced debonding, or formation washouts in fractured volcanic lithologies. In such scenarios, MEPs can serve as a remedial zonal isolation device to mitigate interzonal crossflow and thermally induced casing deformation. This capability is critical for sealing off cold zones that might otherwise compromise well productivity.

The system's adaptability, coupled with thorough diagnostics and design optimization, positions the MEP as a technically viable and field-proven solution for long-section relining in high-risk and legacy geothermal wells.

Field Results and Diagnostic Interpretation

Following several months of detailed engineering design, operational planning and cross-disciplinary coordination, total of four geothermal production and one injection wells have been successfully relined in the span of almost a year using 1214MEP and 812MEP. The former being mounted on a 9-5/8 in. casing and set inside a 13-3/8 in. casing and the latter being mounted on a 7 in. casing and set inside 9-5/8 in. casing. These configurations were selected to suit the varying geometries and damage profiles of the wells. All wells exhibited significant casing degradation, including severe internal corrosion, extensive erosion, parted tubulars, and collapsed casing intervals.

Given the high risk of behind-casing channeling, multi-finger caliper logging was conducted prior to intervention to assess internal diameter variations, quantify casing wall thinning, and detect ovality or deformation. This diagnostic data was instrumental in confirming mechanical integrity and identifying optimal setting depths for the MEPs, ensuring that the reline objectives could be achieved for either 13-3/8 in. and 9-5/8 in. production casing strings.

Each sub-assembly comprises of handling pups, a cement port collar, MEP was meticulously measured and precisely spaced out to ensure accurate placement at the predetermined setting depth. This level of precision was critical, as even minor deviations in setting depth could compromise sealing performance, particularly in wells where potential leak across the 13-3/8 in. and 9-5/8 in. production casings for large and regular holes, respectively, due to severe damage.

Upon reaching the target setting depth, a cup or combination tool was run inside the casing to hydraulically set the MEP and subsequently execute the stage cementing operation through the integrated port collar in a single trip. Prior to the cement job, the annulus between the host casing and the reline was pressure tested to approximately 1,000 psi to validate that the MEP had fully expanded and was capable of sustaining the hydrostatic pressure of the cement slurry during displacement and curing. This validation step was essential to ensure cement containment, eliminate the risk of crossflow, and prevent interzonal fluid migration during or after the cement curing period.

In addition to its application in well remediation, the same MEP configuration has proven effective in new wells, particularly in scenarios where a cold zone is identified below the shoe of a previous casing string and cannot be isolated using conventional casing programs. In such cases, the same MEP can be deployed and set in open hole section, highlighting its flexibility and ability to conform to irregular borehole geometries which allows for effective sealing against formation washouts or fractured volcanic lithologies.

This capability provides a proactive solution for zonal isolation or temperature and pressure interference that could compromise well performance. The rig time savings achieved by eliminating the need for a dedicated drill-out run and minimizing operational trips further enhanced the efficiency and economic value of the reline approach.

By integrating expandable packer technology into both workover and drilling strategies, geothermal operators can move away from reactive approach and toward a more proactive, design-driven approach to well construction and long-term integrity management. This shift not only extends the operational lifespan of geothermal wells but also contributes to a reduction in the number of wells required per project, ultimately lowering the Levelized Cost of Energy (LCOE) for geothermal developments. The result is a more sustainable, reliable, and economically viable renewable energy investment profile, supporting broader energy transition goals.

Conclusions and Recommendations

Historically, the geothermal drilling industry has predominantly relied on drillable plugs as a conventional solution for relining operations. However, these approaches have often introduced additional mechanical stress to already compromised casings and contributed to a reduction in overall life of the well. The limited versatility of such systems constrained strategic choices for drilling and well intervention engineers, particularly in high-temperature and chemically aggressive geothermal environments.

This paper demonstrated the successful application and potential of a novel metal expandable packer (MEP), featuring a flexible and innovative design tailored for both cased and open hole geothermal applications. The MEP serves as a pressure-tight barrier, effectively mitigating cement contamination risks during stage cementing. When used in conjunction with an integrated cement port collar, this system eliminates the need for dedicated drill-out runs, thereby restoring well integrity while offering substantial rig time savings and enabling faster well reactivation for production or reinjection.

The integration of bespoke valve block configuration further extends the applicability of the MEP to long reline intervals without requiring multiple packers and ensure that the production casing is relined from top of the liner to surface. This provides both operational flexibility and economic efficiency during initial well construction and future workovers, making it particularly advantageous in complex geothermal field developments.

Despite the presence of thermal cycling stresses and chemically aggressive downhole environments, the MEP's robust architecture, supported by a diverse range of seal materials and valve block configurations, enables reliable performance in relining and tieback applications. This is especially valuable in low pH geothermal wells, where conventional sealing technologies frequently fail due to accelerated corrosion and material degradation. The MEP's high-temperature anchoring capability and mechanical resilience position it as a viable long-term integrity solution for demanding geothermal operations.

Furthermore, this technology has proven, with its 100% service quality to date for this campaign that a specially engineered technology offers a strategic alternative to traditional remedial practices, aligning with industry goals of prolonging well life, minimizing intervention frequency, and enhancing resource production. As such, the MEP-based solution represents a significant advancement in geothermal well construction and workover methodologies, with the potential to improve safety, operational repeatability, and lifecycle control in geothermal asset integrity management.

By integrating this high-performance MEP into geothermal workovers, operators gain a technically validated method to extend well life, reduce intervention costs, and re-establish zonal integrity in wells that would otherwise be abandoned or require expensive redrilling. It represents a critical advancement in geothermal well remediation precisely engineered for the high-temperature, high-stakes environments of modern geothermal energy development.

At the time of writing, field deployments of more MEPs are still ongoing, with regular performance monitoring to evaluate long-term reliability under representative geothermal field conditions.

Nomenclature

CRA	- Corrosion-Resistant Alloy
in.	- Inch
LCOE	- Levelized Cost of Energy
MD	- Measured Depth
mMD	- Meter Measured Depth
MEP	- Metal Expandable Packer
psi	- Pounds per Square Inch
PTS Survey	- Pressure-temperature-spinner (PTS) Survey
SCP	- Sustained Casing Pressure
°C	- Degree Celsius

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